

SafeConf supplement

Supplement draft index

S1. Information and split contracts

Task identities, gene-cluster splits, public history eligibility, source-error lineage, target-feedback budgets, and frozen evaluation time order.

S2. Public evidence baselines

Magnitude, support, history energy, direct distance, PublicRule, missing-history fallback, and support-matched content permutations.

S3. Feedback and PertEMA adaptation

Native control features, Public additions, target error budgets, source/target error CDFs, fold-out score generation, and the distinction between the current adaptation and the complete official PertEMA conformal workflow.

S4. Cross-family stress test

Frangieh GEARS↔scGPT direction tables, 15 risk strata, paired gene-cluster bootstrap, and the six failed upstream competence gates.

S5. KOLF frozen independent evaluation

Role registry, 1,400-axis contract, 1,366-coordinate public adapter, predictor gate, risk-score hash, coverage report, no-history fallback check, and frozen evaluation metrics.

S6. Negative and boundary results

DeepSets and pointwise public-set models, background mapping, Source gate, Target label-equivalent budget right-censoring, content attribution, and KOLF broad external intervals.

SafeConf supplement (continued)

S7. Reproduction

See ``evidence_freeze_v1/REPRODUCTION.md``; every figure has source data and every external result has a hash-bound artifact.